

EAFM-Formula Sheet 1

$D = \frac{\left(\sum_{t=1}^T PV(CF_t) \times t \right)}{P_0}$
$D_{perpetuity} = \frac{(1+y)}{y}$
$D_{annuity} = \frac{1+y}{y} - \frac{T}{(1+y)^T - 1}$
$D = \frac{1+y}{y} - \frac{(1+y) + T(c-y)}{c[(1+y)^T - 1] + y}$
$D_{parbond} = \frac{1+y}{y} \left[1 - \frac{1}{(1+y)^T} \right]$
$FVIFA_{ordinary} = \frac{[(1+r)^n - 1]}{r}$
$FVIFA_{due} = \frac{[(1+r)^n - 1] \times (1+r)}{r}$
$PVIFA_{ordinary} = \frac{\left[1 - \frac{1}{(1+r)^n} \right]}{r}$
$PVIFA_{due} = \left[\frac{\left[1 - \frac{1}{(1+r)^n} \right]}{r} \right] \times (1+r)$
$Convexity = \frac{(PV_-) + (PV_+) - 2 \times (PV_0)}{(\Delta yield)^2 \times PV_0}$
$D^* = \frac{(PV_-) - (PV_+)}{2 \times (\Delta Yield) \times (PV_0)}$
$i_{1,H} = i_{1,L} e^{2\sigma} \quad \text{effective } \square$
$i_{1,H} = i_{1,L} e^{2\sigma} \quad \text{tree}$

$$F_{(t^*, T)} = \frac{(1+r_t)^t}{(1+r_{t-1})^{t-1}} - 1$$

$$FR_t = \frac{(1+z_t)^t}{(1+z_{t-1})^{t-1}} - 1$$

z_t = zero swap rate

$$FR_t = \left[\frac{(1 + \frac{z_t}{4})^t}{(1 + \frac{z_{t-1}}{4})^{t-1}} - 1 \right] \times 4 \quad (\text{annualized})$$

$$AIP_{FRB} = \sum_t \left[\frac{C_t}{(1+r_t)^t} \right] + \frac{FV}{(1+r_t)^n}$$

$$r_t = \frac{z_t + TS}{4}$$

$$C_t = \frac{JIBAR + IS}{4} \times FV$$

$$C_r = \left[\frac{(1 + \frac{r_n}{4})^n}{(1 + \frac{r_{n-1}}{4})^{n-1}} - 1 \right] \times 4$$

$$AI_{FRB} = \text{Annual } C @ NCD \times \frac{LCD - SD}{365} \quad \text{cum-coup}$$

$$AI_{FRB} = AC @ NCD \times \frac{SD - NCD}{365} \quad \text{ex-coup}$$

$$CPI \text{ Index ratio} = \frac{CPI_{SD}}{CPI_{ID}}$$

$$REFCPI_{ID} = \frac{CPI_{t+1} - CPI_t}{D} \times (t-1) + CPI_t$$

$$AIP = AIP \times CPI \text{ Index}$$

$$\frac{\Delta p}{p} \approx -D^* \times \Delta y$$

$$\frac{\Delta p}{p} \approx -D^* \times \Delta y + \left(\frac{1}{2} \times \text{conv} \times \Delta y^2 \right)$$

